



PIYUSH MAJI
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EDUCATION

Indian Institute of Technology Hyderabad
M.Sc. in Physics CGPA: 7.5

Expected July 2027

Hooghly Mohsin College (1836), West Bengal
B.Sc. in Physics CGPA: 7.28

August 2024

RESEARCH

Graph-based Full Event Interpretation (GraFEI) | Belle II Experiment

Ongoing

Python • PyTorch Geometric • Graph Neural Networks • basf2

- Addressed the challenge of reconstructing complex B -meson decay topologies without predefined decay-chain assumptions by developing an end-to-end **Graph Neural Network (GNN)** pipeline that learns directly from detector-level particle observations.
- Engineered fully connected **graph representations** using particle kinematics and detector observables (DOCA, $\cos \theta$, momentum, charge), and implemented a **multi-task GNN** in **PyTorch Geometric** to jointly predict **particle identities** and **hierarchical decay structures (LCAS)**.
- Validated the model within the **Belle II basf2** framework using **PerfectLCA**, **PerfectMasses**, and **PerfectEvent** metrics, contributing to **ML-driven event reconstruction** for rare **high-energy physics** analyses.

PROJECTS

GNN-FraudNet | Graph-based Financial Fraud Detection

Completed

Python • PyTorch Geometric • FastAPI • XGBoost • Docker [GitHub](#)

- Addressed the challenge of detecting **fraudulent Bitcoin transactions**, where traditional ML models ignore transaction relationships, by developing a **Graph Neural Network** pipeline that leverages graph connectivity for node classification.
- Engineered a graph learning workflow on the **Elliptic Bitcoin Dataset (203K+ transactions, 234K+ edges)** using **GraphSAGE** and **Graph Attention Networks (GAT)**, with preprocessing, class balancing, and explainability through **SHAP** and **GNNExplainer**.
- Built a production-ready inference pipeline by exposing fraud predictions through a **FastAPI** REST API and containerizing the application with **Docker** for scalable deployment.

Retail Demand Forecasting & Inventory Optimization

Completed

Python • LightGBM • Time Series Forecasting [GitHub](#)

- Solved the problem of **retail demand forecasting** by building a scalable **SKU-level sales prediction pipeline** using **LightGBM**, enabling accurate inventory planning across multiple stores.
- Designed advanced time-series features including **lag**, **rolling-window**, **calendar**, **promotion**, and **holiday** variables, and benchmarked the model against **SARIMA** and **Prophet** baselines using time-aware cross-validation.
- Implemented **hierarchical forecast reconciliation** to maintain consistency across **SKU**, **store**, and **regional forecasts**, improving the usability of predictions for supply chain and inventory decision-making.

Spam Classification from Scratch

Completed

Python • NumPy • Machine Learning [GitHub](#)

- Built a machine learning pipeline from scratch to **detect spam SMS messages**, implementing **Multinomial Naive Bayes** and **Logistic Regression** without using **scikit-learn** or other ML libraries.
- Implemented custom **Bag-of-Words**, **TF-IDF**, **MLE**, **Laplace smoothing**, **gradient descent**, and end-to-end evaluation metrics (**ROC**, **AUC**, **F1**) while ensuring **data-leakage-free** feature engineering.
- Achieved **94.9% F1-score** and **0.982 AUC** on the **UCI SMS Spam Collection**, demonstrating strong performance through threshold optimization and comparative model analysis.

POSITIONS OF RESPONSIBILITY

Placement Coordinator, Office of Career Services, IIT Hyderabad

May 2026 – Present

Manage end-to-end campus recruitment drives and liaise with recruiters; onboarded companies including **IBM**, **Licious**, **HCL**.

Inbound & Logistics, International Relations Cell, IIT Hyderabad

Sep 2025 – Present

Coordinate orientations, cultural programs, and academic tours for **exchange students from Japan**, **international delegates**, and **company officials**.

Design Core, Extra Mural Lectures (EML), IIT Hyderabad

May 2026 – Present

Lead visual design and event branding for EML; hosted **NASA Axiom Mission Pilot Subhanshu Shukla** for a campus lecture.

SKILLS

Languages & Tools: Python, SQL, Git, NumPy, Pandas, PyTorch, PyTorch Geometric, LightGBM, basf2

Domain: Machine Learning, Graph Neural Networks, A/B Testing & Experimentation, Time Series Forecasting, Data Science, Data Structures & Algorithms

Coding Practice: LeetCode, GeeksforGeeks, Striver A2Z, Codeforces

RELEVANT COURSES

Machine Learning for Physical Sciences • Machine Learning for High Energy Physics • Statistical Mechanics • Linear Algebra • Computational Physics • Data Science Analysis